

Unit 9, Lesson 1a: Area of a Polygon



Benchmark

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.



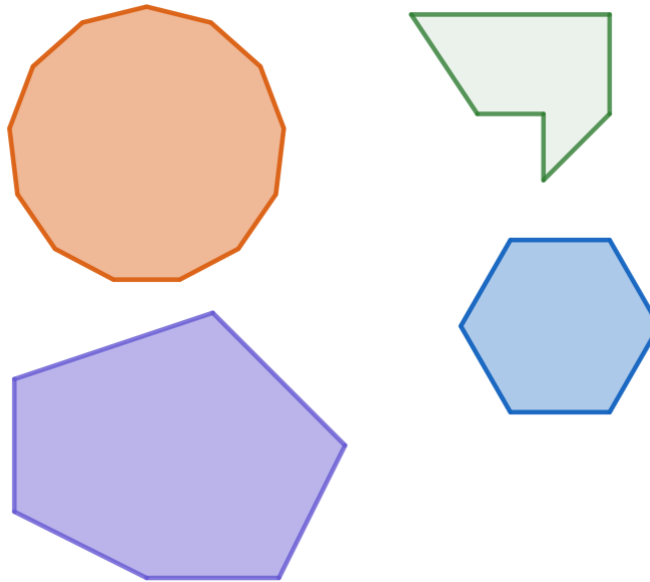
Learning Target

- ☐ I can solve mathematical and real-world problems involving the area of a polygon.



9.1a.1 Warm-Up: Which One Doesn't Belong?

Four different polygons are shown.



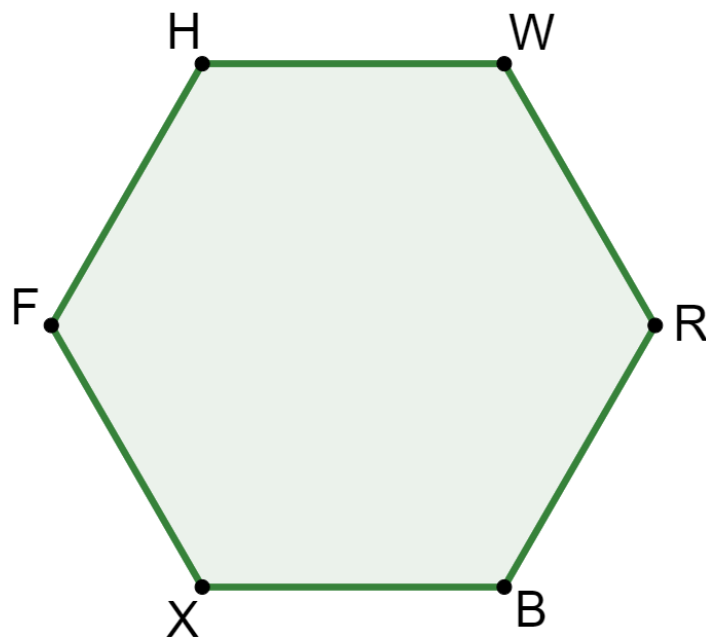
- Which polygon does not belong?
Answers will vary.
- What characteristic did you use to make your decision?
Answers will vary. Teachers may wish to survey the class to determine which characteristics were used to make a decision.



9.1a.2 Collaboration: Decomposing a Regular Polygon

Teacher Notes: This collaboration is designed to help students consider the dissection of a regular polygon into congruent triangles and information that is needed to find the area of the polygon when polygons are dissected into triangles. As you circulate, avoid giving students correct answers but listen for possible misconceptions or for prior knowledge that may need support during the guided instruction.

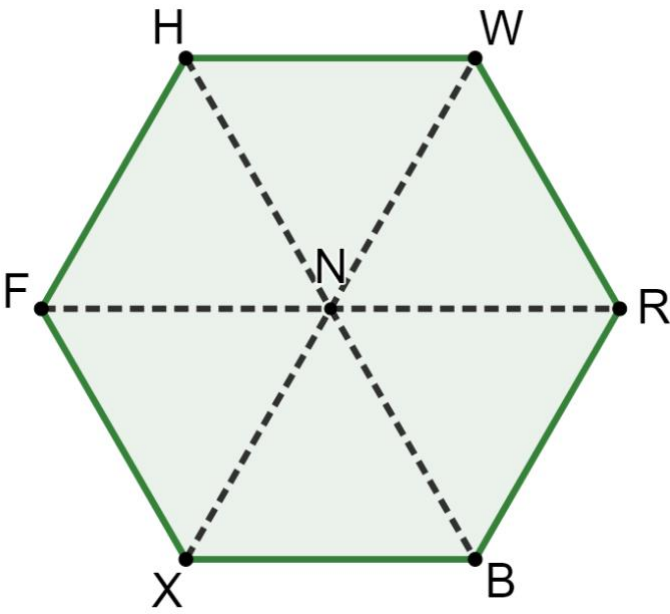
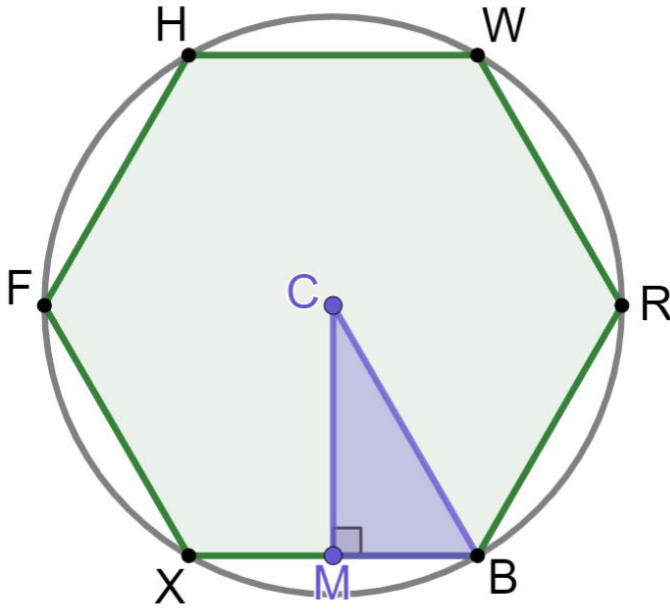
Regular polygon $HWRBXF$ is shown.



1. Discuss with your partner what information is necessary to determine the area of the polygon. Summarize your discussion.

Answers will vary. Allow for student autonomy.

2. Gregg and Dariq shared their ideas with their class.

Gregg's Idea	Dariq's Idea
	
I drew the diagonals to find the center of the polygon. This resulted in 6 congruent triangles.	I found the center of the polygon by using a compass to draw a circle that passes through the vertices. I then found the midpoint of one side and drew a right triangle.

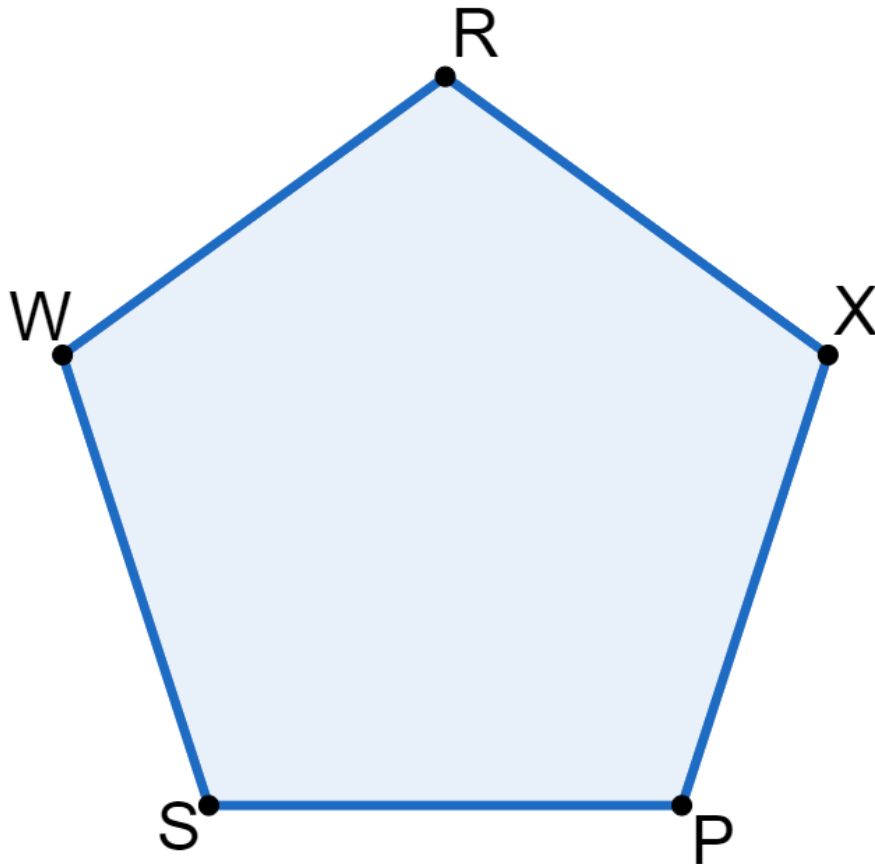
- a. Determine what, if any, further information Gregg needs to determine the area of the polygon.

Answers will vary. This method is demonstrated to help students see the congruent triangles so that a foundation is built for the development of the formula.

- b. Determine what, if any, further information Dariq needs to determine the area of the polygon.

Answers will vary. This method is used to help students relate the center of a circle to the center of a polygon. It is also used to relate the apothem as the side of a right triangle.

3. Regular polygon $RXPWS$ is shown.



Determine whether Gregg's or Dariq's method could be used to determine the area of $RXPWS$. Explain your answer or show the method on the diagram.

Answers will vary. As you circulate, you may wish to gather information from students to share with the class. Look for students who use the perpendicular bisectors of each side. This is another good connection for students to make. If desired, use <https://www.geogebra.org/geometry/pvh3ryv5> to construct the perpendicular bisectors for the class and ask the students what the point of intersection could be. Then, draw the circumscribed circle.



9.1a.3 Exploration: Regular Polygons

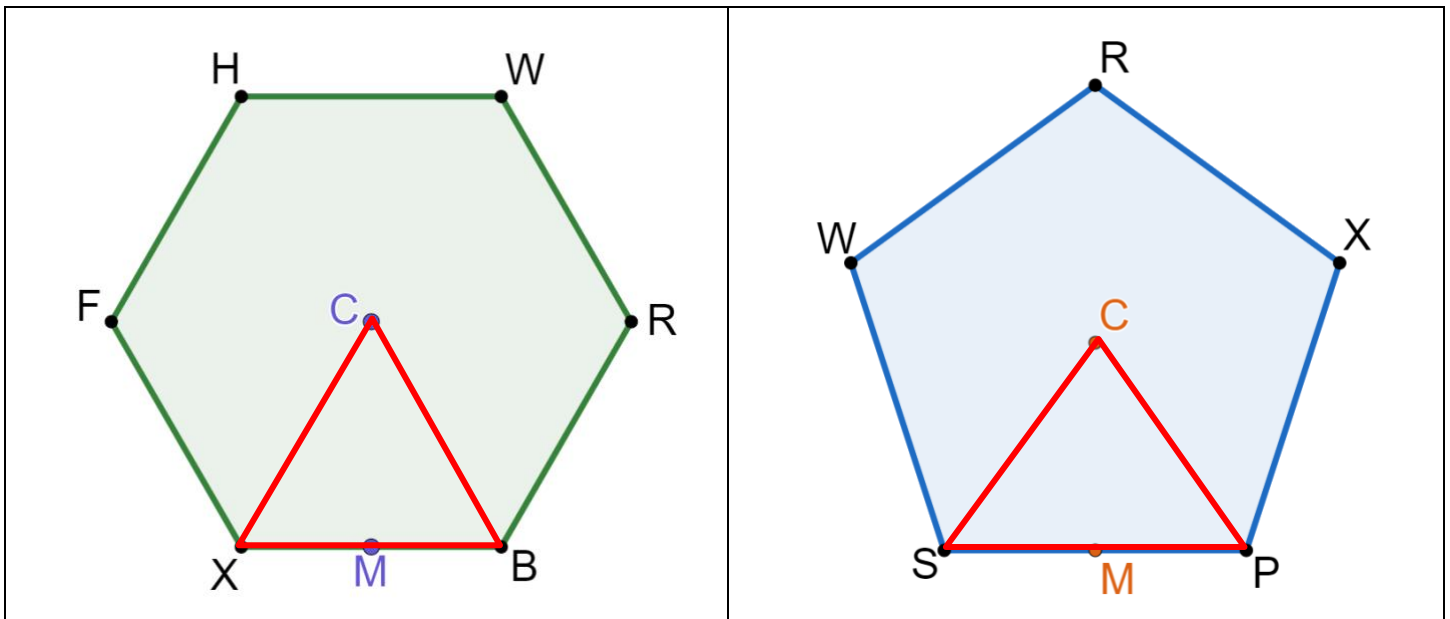


Regular Polygon

A polygon that is both equilateral (all sides congruent) and equiangular (all angles congruent).

Teacher Notes: This exploration builds the connection between the area of triangles to the area of the polygon. Rather than just giving the formula, this exploration has the students relate the pieces of the formula to decomposing the polygon into triangles with the center as a common vertex. Once students complete the exploration, then ensure students agree on the answers to the statements in question 2.

Polygons $HWRB XF$ and $RXPSW$ are both regular polygons. The diagram shows the center of each polygon and the midpoint of one side.



1. In polygon $HWRB XF$, draw ΔXCB , and in polygon $RXPSW$, draw ΔSCP .

2. Complete the statements.

The area of $\triangle XCB$ is equal to one-half times the length of

- ☐ $\overline{XM}$
- ☒ $\overline{XB}$

times the length of

- ☒ $\overline{CM}$.
- ☐ $\overline{CX}$.

The area of polygon $HWRBXF$ can be

found by multiplying the area of $\triangle XCB$ by

- ☐ 4
- ☐ 5
- ☒ 6
- ☐ 7

because polygon

$HWRBXF$ can be decomposed into

- ☐ 4
- ☐ 5
- ☒ 6
- ☐ 7

triangles that are congruent

to $\triangle XCB$. The area of $\triangle SCP$ is equal to one-half times the length of

- ☐ $\overline{SM}$
- ☒ $\overline{SP}$

times the length of

- ☒ $\overline{CM}$.
- ☐ $\overline{CS}$.

The area of polygon

$RXPSW$ can be found by multiplying the area of $\triangle SCP$ by

- ☐ 4
- ☒ 5
- ☐ 6
- ☐ 7

because polygon $RXPSW$ can be decomposed into

- ☐ 4
- ☒ 5
- ☐ 6
- ☐ 7

triangles

that are congruent to $\triangle SCP$. The number of triangles in polygon

$HWRBXF$

- ☒ is
- ☐ is not

equal to the number of sides. The number of

triangles in polygon $RXPSW$

- ☒ is
- ☐ is not

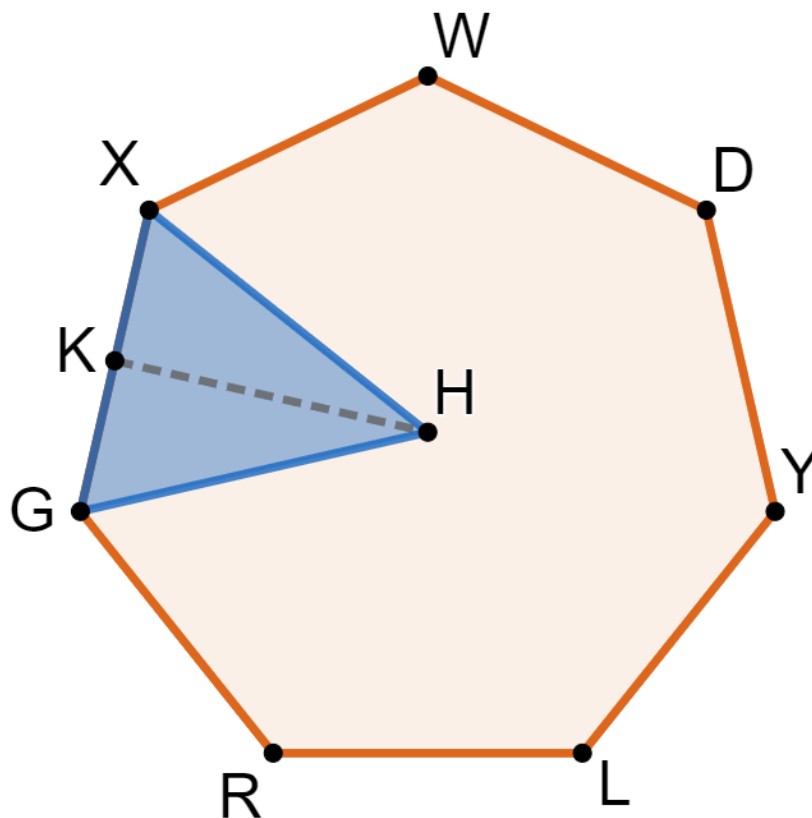
equal to the number of sides.



9.1a.4 Guided Instruction: Area of a Regular Polygon

Teacher Notes: This guided instruction is meant to guide students to understand what part of a polygon is an apothem. Allow students opportunity to write their own definition so that the word has meaning for them. The instruction also leads the student to the formula.

Regular polygon $XWDYLRG$ is shown, where $\overline{KH} \perp \overline{XG}$, $\overline{XK} \cong \overline{KG}$, $XG = 6$, and $KH = 6.23$.



1. Determine the area of $\triangle XHG$.
18.69
2. Use the information to determine the area of polygon $XWDYLRG$.
130.83
3. Determine the value of the expression $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$.
130.83

4. What do you notice about the value of $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$ and the area of polygon $XWDYLRG$?
The expression gives the same value as the area of the polygon. Teachers may wish to ask students why they think the values are the same.
5. In the expression, $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$, what does the 7 represent?
The polygon has 7 sides.
6. What part of polygon $XWDYLRG$ does $7 \cdot XG$ represent?
the perimeter
Students may need some probing to reach this answer.

The formula for finding the area of a regular polygon is $A = \frac{1}{2}Pa$, where P is the perimeter of the polygon and a is the apothem.

7. For polygon $XWDYLRG$, the expression $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$ was used. Discuss with your partner which part of the expression $\frac{1}{2} \cdot 7 \cdot XG \cdot KH$ is the apothem.
Teachers may wish to circulate while students discuss. Allow students to grapple with how to word the definition.
8. Write a definition of an apothem.
Answers will vary.

Share your definition of an apothem with a classmate by stating the definition to each other. Record your classmate's definition as they state it. *You are the only person who should write in the first two columns of the table.* Have the person initial next to their definition stating that your written definition is correct.

Teacher Notes: Research shows that when a student listens to another student, then their understanding can change. Allow students the time to share. As students progressed through the lesson the apothem was the height of a triangle. Students may choose to define it as such, but point out to students that there are a variety of triangles that could be drawn within a polygon so that definition may not be precise. When writing the class definition, ensure students define the apothem with relation to a regular polygon.

Classmate	Definition of apothem	Initials
	<i>Answers will vary.</i>	

Class definition of apothem.

Answers will vary.

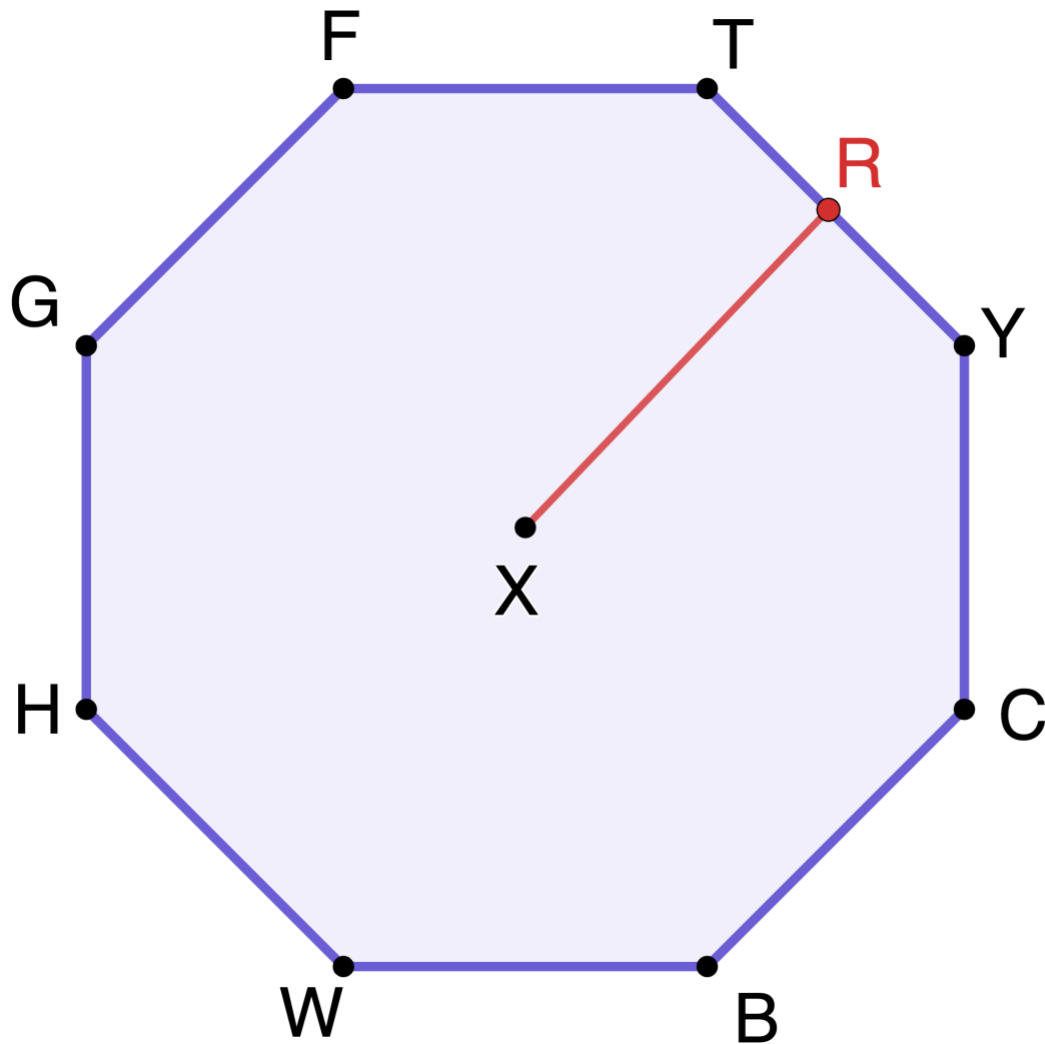
Example: A line segment from the center of a regular polygon to the midpoint of one of its sides.



9.1a.5 Guided Practice: Determining Area of a Polygon

1. A game board is in the shape of regular pentagon where each side measures 45 centimeters (cm) and the length of the apothem is 61.937 cm. Determine the area of the game board. Round your answer to the nearest thousandth.
6967.913 cm²

2. Polygon $FTYCBWHG$ is shown where point X is the center of the polygon, $TX = 65.325$, and $WB = 25$.



- Draw an apothem from point X to TY . Label the point where the apothem meets TY as R .
- Determine the length of TR .
 12.5
- Determine the length of XR . Round to the nearest thousandth.
 64.118
- Determine the area of polygon $FTYCBWHG$. Round to the nearest tenth.
 6411.8



9.1a.6 Your Turn!

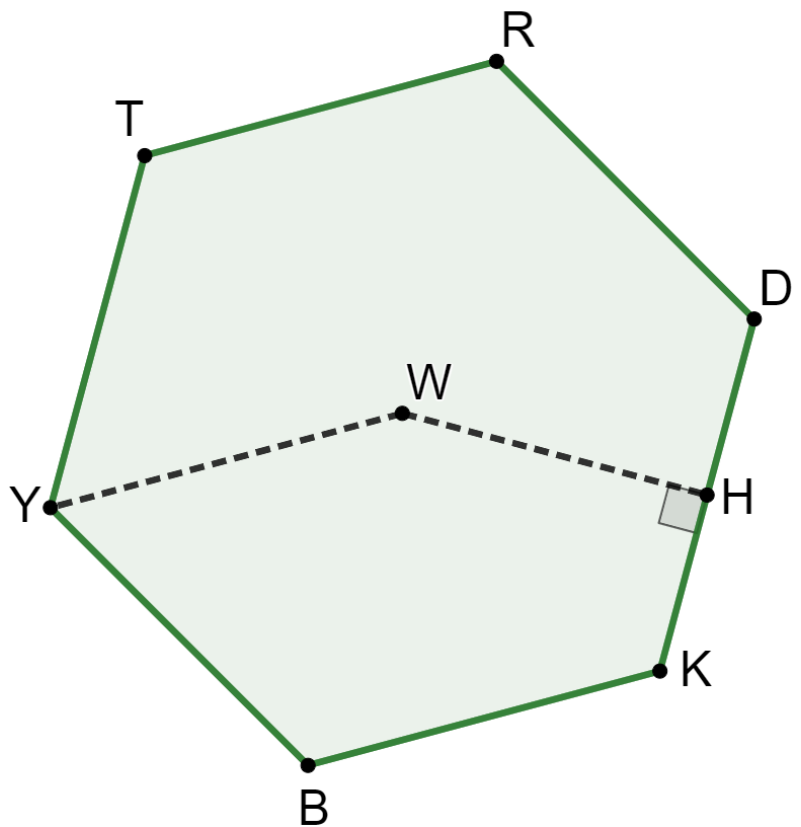
Teacher Notes: Students should not round until the very end. It may be helpful for students to know the difference, although a small difference, between using the rounded value of the apothem to find the area to using the actual value. Instruct students on how to use the actual value when using a calculator.

1. A tile is in the shape of a regular dodecagon (a 12-sided polygon). Each side of the tile measures 74.4 centimeters (cm) and the length of the apothem is 138.83 cm. Determine the area of the tile. Round your answer to the nearest thousandth.

61973.712 cm^2

2. Regular polygon $YTRDKB$ is shown where point W is the center of the polygon, WH is an apothem, $BK = 157$ millimeters (mm), and $WY = 157$ mm. Determine the area of polygon $YTRDKB$. Round your answer to the nearest thousandth.

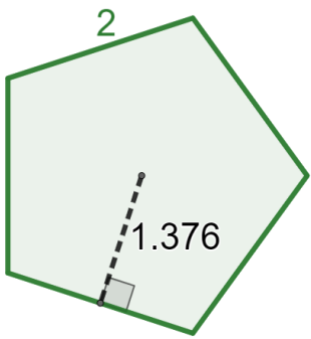
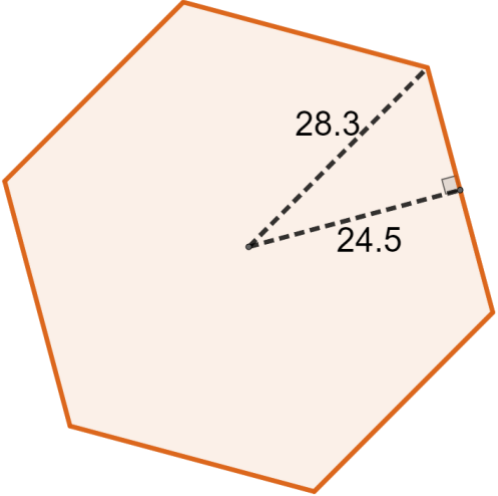
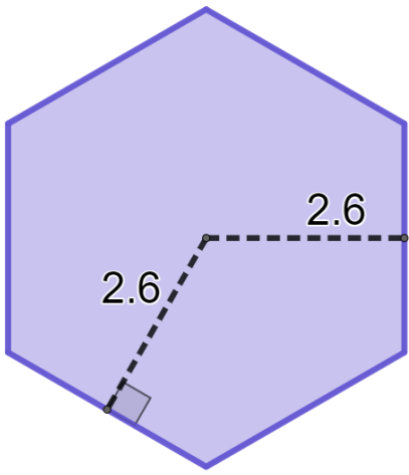
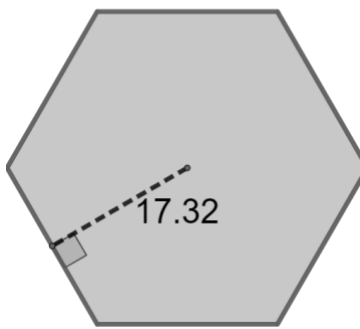
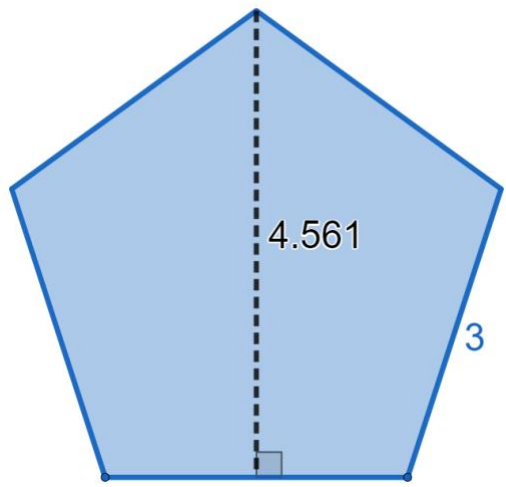
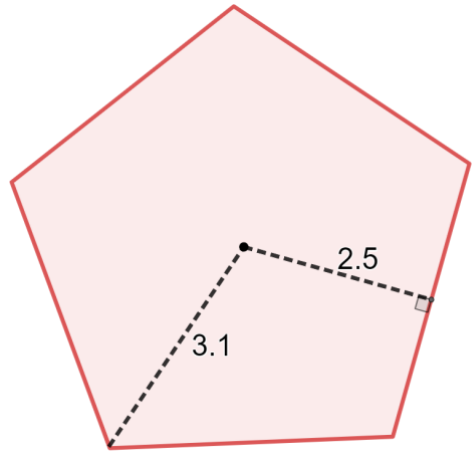
64039.981 mm^2





9.1a.7 Wrap-Up: Enough Information?

1. Six regular polygons are shown with measurements.

Polygon A	Polygon B	Polygon C
 apothem = 1.376	 apothem = 24.5	 apothem = 2.6
Polygon D	Polygon E	Polygon F
 apothem = 17.32	 apothem = 4.561	 apothem = 2.5

Select all of the regular polygons which have enough information to determine the area of the polygon.

- ☐ Polygon A
- ☐ Polygon B
- ☐ Polygon C
- ☐ Polygon D
- ☐ Polygon E
- ☐ Polygon F

Teacher Notes: Teachers may wish to ask students to find the areas of the polygons that have enough information. Some students may choose polygon C stating that the area of the triangle could be found using special right triangles. There is no benchmark in the BEST mathematics benchmarks that teaches the student to find the measure on one angle of a polygon. MA.8.GR.1.6 states "Develop and use formulas for the sums of the interior angles of regular polygons by decomposing them into triangles." This benchmark is prior knowledge but only requires students to determine the sum. Therefore, a student who does this is extending their understanding.